

Artificial Intelligence / Applied Artificial Intelligence

Course Outline

Week No.	Content
1	Introduction to AI Agents, Definitions of AI, History, Current State of the Art
2	Agents + Environments Structure of an Agent, Types of Environments, Types of Agents
3	Local Search Optimization Problems, Hill Climbing, Simulated Annealing
4	Uninformed Searching Depth First Search, Breadth First Search, Uniform Cost Search, Bidirectional Search
5	Informed Search Greedy Search, A* Search, Simple + Recursive Best First Search
6	Sessional I
7	Adversarial Searches Min-Max, Alpha-Beta Pruning
8	Machine Learning 1 Supervised Learning - kNN, Decision Trees/Naive Bayes, Random Forest
9	Supervised Learning Artificial Neural Networks (ANN) - Single layer + Multi-layer
10	Machine Learning 2 Unsupervised Learning - K-means Clustering, PCA, SVMs
11	Multi-layer Perceptron, Backpropagation
12	Sessional II
13	Constraint Satisfaction Problems Graph coloring, N-Queen, etc.
14	Genetic Algorithms Darwin's theory of evolution-based techniques
15	Concepts of Federated Learning, Reinforcement Learning (RL), Ensemble Techniques Brief Introduction and Real-world Applications